

Contents

1	Wang and Hong	2
2	Han et al.	2
3	Shaqour and Hagishima	4
4	Al Sayed et al.	4
5	Yu et al.	6
6	Weinberg et al.	6
7	Latoń et al.	8
8	Fu et al.	8
9	Zhang et al.	10
10	Al-Ani and Das	10
11	Mason and Grijalva	12
12	Huynh et al.	12
13	Kannari et al.	14
14	Chatterjee and Khovalyg	14
15	Sierla et al.	16
16	Pinthurat et al.	16
17	Michailidis et al.	18
18	Li et al.	18
19	Lei et al.	20
20	Mazhar et al.	20
21	Mohsenpour and Xing	22

1 Wang and Hong

Table 1: MRQ1 recoded audit table for Wang and Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
WangHong2020	3	2	The survey characterizes RL methods through algorithm families as one of its main review perspectives. Value-based RL is structurally dominant in the reviewed literature, while policy-gradient and actor–critic methods are discussed as rising families because explicit policy representations may support transfer learning across buildings. Model-based RL is acknowledged but treated cautiously because learning accurate models is time-consuming and expertise-dependent. Recommendation receives score 2 because the survey conditionally supports value-based RL for its practical simplicity and supports policy-gradient/actor–critic methods for transfer-learning potential, but does not present a single universally preferred family.

Table 2: MRQ1 method-family characterization scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Wang and Hong (2020)	1	3	2	2	0

Table 3: MRQ1 method-family recommendation scores for Wang and Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Wang and Hong (2020)	0	2	2	2	0

Table 4: MRQ1 divergence/rationale table for Wang and Hong (2020), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Value-based	Value-based RL is presented as the dominant family in the reviewed building-control literature, especially because Q-learning-style approaches balance implementation simplicity and control performance.	Although dominant historically, value-based RL is not framed as the complete future solution for transfer, generalization, or larger continuous-control settings.
Policy-gradient / policy-based	Policy-gradient methods are discussed as increasingly popular because explicit policy representations may support transfer learning across buildings.	They are not described as the dominant current family and are presented more as a rising direction than as a universal solution.
Actor–critic	Actor–critic methods are also discussed as increasingly popular and potentially useful for transfer learning and more complex control settings.	The survey does not show actor–critic methods as clearly superior overall; support remains conditional and task-dependent.
Model-based / hybrid	Model-based RL is acknowledged as a valid method family and is compared conceptually with MPC-like approaches.	The survey is cautious because learning accurate models can be time-consuming, expertise-dependent, and not always beneficial for control performance.

Table 5: One-line MRQ1 answer for Wang and Hong (2020), after recoding.

survey_id	MRQ1 answer
WangHong2020	Wang and Hong (2020) portray value-based RL as the historically dominant family in building controls, while policy-gradient and actor–critic methods are framed as rising families because of their transfer-learning and generalization potential; model-based RL is acknowledged but treated cautiously because of modeling burden.

2 Han et al.

Table 6: MRQ1 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
HanEtAl2021	3	2	The survey structurally discusses value-based, policy-based, and actor–critic reinforcement-learning methods in the context of occupant behavior in building control. Value-based and tabular methods are historically dominant and are described as suitable for small and discrete problems because of simplicity, convergence, and robustness. However, the survey also states that classical tabular Q-learning becomes insufficient for stochastic and complex occupant-behavior control, especially under continuous or high-dimensional state spaces. Policy-based methods are discussed as suitable for larger or continuous problems, while actor–critic is presented as an alternative that still needs more applications. Recommendation receives score 2 because the survey gives problem-dependent guidance rather than one universal preferred method family.

Table 7: MRQ1 method-family characterization scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor-critic	Multi-agent / distributed
Han et al. (2021)	0	3	2	2	0

Table 8: MRQ1 method-family recommendation scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor-critic	Multi-agent / distributed
Han et al. (2021)	0	2	2	1	0

Table 9: MRQ1 divergence/rationale table for Han et al. (2021), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Value-based	Value-based methods are easy to implement, can guarantee convergence in tabular settings, are robust, and remain the most common family in reviewed occupant-behavior RL studies. They are conditionally suitable for small and discrete state/action problems.	Classical tabular Q-learning becomes insufficient for stochastic, complex, continuous, or high-dimensional occupant-behavior problems, where approximation or richer representations are needed.
Policy-gradient / policy-based	Policy-based methods are conditionally supported for large MDPs and continuous-state problems because they avoid explicit tabular value representation and can directly represent policies.	They are not framed as universally best; practical implementation and wider empirical evidence remain less developed than for value-based methods.
Actor-critic	Actor-critic methods are presented as a distinct alternative for more complex settings.	The survey explicitly treats actor-critic as less mature in this application area and states that more applications need to be developed.

Table 10: MRQ1 strategy-level notes for Han et al. (2021), after recoding.

survey_id	note_type	note
HanEtAl2021	Training strategy	Off-policy methods are favored for complex control tasks because learning becomes more efficient when exploration and repeated real signal collection are difficult.
HanEtAl2021	Specific algorithm guidance	SARSA may be preferred when the agent is exploring in error-insensitive systems.
HanEtAl2021	Specific algorithm guidance	Q-learning is preferred when agents do not explore.

Table 11: One-line MRQ1 answer for Han et al. (2021), after recoding.

survey_id	MRQ1 answer
HanEtAl2021	Han et al. (2021) present a problem-dependent algorithm narrative: value-based methods remain historically dominant and suitable for small discrete problems, but classical tabular Q-learning is increasingly insufficient for stochastic and complex occupant-behavior control; policy-based methods are conditionally supported for larger or continuous problems, while actor-critic is treated as a promising but less mature alternative.

3 Shaqour and Hagishima

Table 12: MRQ1 recoded audit table for Shaqour and Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
ShaqourHagishima2022	3	3	The survey structurally characterizes DRL algorithm families across different building types using extensive algorithm tables and building-type-specific comparisons. Value-based methods such as DQN, DDQN, BDQ, and fitted Q-iteration are repeatedly visible and remain important, especially in residential studies. Actor–critic methods such as DDPG, A3C, A2C, PPO, SAC, TD3, ACKTR, and MAAC are strongly represented across residential, office, and data-center contexts. Model-based or hybrid methods such as RLMPC and model-based DRL are acknowledged but less central. Multi-agent and distributed DRL methods are structurally emphasized and explicitly recommended for future neighbourhood- and multi-building energy management. Recommendation receives score 3 because the survey gives clear future-oriented support for multi-agent/distributed DRL and also explicitly recommends SAC for future utilization in data-center HVAC.

Table 13: MRQ1 method-family characterization scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Shaqour and Hagishima (2022)	2	3	0	3	3

Table 14: MRQ1 method-family recommendation scores for Shaqour and Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Shaqour and Hagishima (2022)	1	2	0	3	3

Table 15: MRQ1 divergence/rationale table for Shaqour and Hagishima (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	DQN and related value-based methods are repeatedly visible in the reviewed literature and remain practically important, especially in residential building studies.	The survey does not elevate value-based methods as the single best family across all building types; later and more complex contexts show diversification toward actor–critic, multi-agent, and hybrid methods.
Actor–critic	DDPG, A3C, A2C, PPO, SAC, TD3, ACKTR, and MAAC are strongly represented across building-type tables; SAC is explicitly recommended for future utilization in data-center HVAC because of its sample efficiency in noisy environments.	The survey still presents trade-offs across algorithms and building types rather than a single universally best actor–critic method.
Model-based / hybrid	Hybrid methods such as model-based DRL and RLMPC are presented as useful combinations that can leverage multiple method families.	Model-based/hybrid methods are not presented as the main overall recommendation or dominant future direction.
Multi-agent / distributed	Multi-agent and distributed approaches are explicitly framed as key future methods for neighbourhoods, multi-building settings, and coordinated energy management.	The strongest support is future-oriented and scaling-oriented rather than a universal recommendation for every building type or task.

Table 16: One-line MRQ1 answer for Shaqour and Hagishima (2022), after recoding.

survey_id	MRQ1 answer
ShaqourHagishima2022	Shaqour and Hagishima (2022) present a building-type- and application-dependent algorithm narrative rather than a single dominant winner: value-based methods remain prominent through DQN/DDQN-style approaches, actor–critic methods are strongly supported through DDPG, A2C/A3C, PPO, ACKTR, SAC, TD3, and related algorithms, and the clearest future-oriented recommendation is distributed and multi-agent DRL for neighbourhood- and multi-building energy management.

4 Al Sayed et al.

Table 17: MRQ1 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
AlSayedEtAl2024	3	2	The survey structurally characterizes reinforcement-learning algorithm families for HVAC control through a dedicated classification section and an implementation overview. Value-based methods, especially DQN, are presented as the current leading trend and are linked to discrete action-space control. Actor–critic methods are also structurally important, with DDPG and SAC identified as leading algorithms and with PPO/SAC examples showing strong HVAC-control performance. Policy-based methods are included in the theoretical taxonomy, but the paper states that no algorithms from this category have yet been implemented in BEMS. Model-based methods are less common in the reviewed literature, but are discussed as sample-efficient and potentially useful for reducing costly real-world interactions. Recommendation receives score 2 because the paper gives conditional support to value-based and actor–critic approaches as current practice, while future-oriented support is stronger for model-based RL and meta-RL as ways to improve sample efficiency, adaptation, and real-building readiness.

Table 18: MRQ1 method-family characterization scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Al Sayed et al. (2024)	3	2	3	2	0

Table 19: MRQ1 method-family recommendation scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Al Sayed et al. (2024)	2	0	2	2	0

Table 20: MRQ1 divergence/rationale table for Al Sayed et al. (2024), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based methods are described as the current leading trend in the HVAC RL literature, with DQN identified as the dominant algorithm. They are linked to practical use in discrete action-space control.	The paper’s future-direction logic does not simply endorse continuing with current dominant methods; it stresses that real-world deployment, adaptation, and generalization remain unresolved.
Actor–critic	Actor–critic methods form a significant part of the literature, with DDPG and SAC identified as leading algorithms. The paper also reports examples where PPO or SAC-style methods improve HVAC-control performance compared with simpler baselines.	Actor–critic methods are not presented as universally best; the paper stresses broader needs such as better generalization, safety, and real-building adaptability.
Policy-gradient / policy-based	Policy-based methods are included in the theoretical RL taxonomy and therefore are part of the method-family characterization.	The paper explicitly states that this category has not yet been implemented in BEMS, so it receives no clear practical recommendation.
Model-based / hybrid	Model-based RL is discussed as sample-efficient and potentially useful because synthetic samples or learned models can reduce costly real-world interactions.	Model-based controllers make up only a smaller portion of the reviewed literature, and accurate environment modeling remains difficult.
Meta-RL	Meta-RL is highlighted as an up-and-coming solution for improving generalization and reducing the computational cost of adaptation before real deployment.	Meta-RL is kept as a future-direction note rather than a main family column because it is discussed mainly as a generalization/adaptation strategy.

Table 21: MRQ1 future-direction notes for Al Sayed et al. (2024), after recoding.

survey_id	note_type	note
AlSayedEtAl2024	Future direction	Meta-RL is presented as an up-and-coming solution for improving generalization and reducing the computational cost of adaptation before real deployment.
AlSayedEtAl2024	Future direction	Future work should focus on real-building deployment by addressing poor diversification of exogenous state components during training.
AlSayedEtAl2024	Future direction	Model-based RL is presented as potentially beneficial for learning from limited interactions through synthetic samples or learned environment models.

Table 22: One-line MRQ1 answer for Al Sayed et al. (2024), after recoding.

survey_id	MRQ1 answer
AlSayedEtAl2024	Al Sayed et al. (2024) characterize the current HVAC RL literature as led by value-based methods, especially DQN, with actor–critic methods forming a substantial second tier led by DDPG and SAC, while pure policy-based methods remain practically absent; the paper’s clearest future-oriented recommendation is to improve generalization and real-building adaptability, with model-based RL and Meta-RL highlighted as promising routes.

Table 23: MRQ1 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
YuEtAl2021	3	3	The survey structurally classifies DRL methods for smart-building energy management and gives representative algorithms across value-based, policy-based, model-based, and multi-agent families. Value-based DRL, including DQN and its variants, is treated as a core family for discrete-action SBEM, while policy-based or actor–critic-style DRL methods such as DDPG, PPO, and A2C/A3C are discussed as representative algorithms for continuous control, stable learning, and reliable training. Model-based DRL is treated as a major family because it can improve sample complexity by learning or using environment models. Multi-agent DRL is structurally important because the paper frames multi-zone, multi-building, and microgrid-level SBEM as scalability problems that require multi-agent formulations. Recommendation receives score 3 because the paper gives strong future-oriented support to model-based DRL for sample efficiency and to multi-agent DRL for scalable practical large-building or multi-building control, while giving conditional support to value-based and policy-based methods depending on action-space structure and learning requirements.

Table 24: MRQ1 method-family characterization scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Yu et al. (2021)	3	3	3	3	3

Table 25: MRQ1 method-family recommendation scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Yu et al. (2021)	2	2	2	3	3

Table 26: MRQ1 divergence/rationale table for Yu et al. (2021), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	DQN and its variants are presented as representative model-free DRL algorithms for SBEM. Experience replay can improve data efficiency, and value-based methods remain natural for discrete action-space problems.	Value-based methods are limited by discrete action support and risks such as Q-value overestimation or suboptimal policies in more complex settings.
Policy-gradient / policy-based	Policy-based and related methods are presented as representative DRL options for SBEM, especially where continuous action spaces, stable learning, or reliable training are important.	Different algorithms have different trade-offs; some may be sensitive to hyperparameters, architecture, or learning instability.
Actor–critic	Actor–critic-style algorithms such as DDPG, PPO, and A2C/A3C are discussed as representative methods for continuous-control and practical SBEM tasks.	The paper does not present one actor–critic method as universally superior; algorithm suitability depends on action-space structure and system complexity.
Model-based / hybrid	Model-based DRL is strongly supported because it can outperform model-free algorithms in sample complexity and can use learned or calibrated environment models to generate future episodes.	Accurate building-environment modeling is difficult, which limits practical applicability.
Multi-agent / distributed	Multi-agent DRL is strongly supported for scalable multi-zone, multi-building, and microgrid-level SBEM. The paper explicitly formulates large-scale SBEM as a Markov-game problem and expects more scalable multi-agent DRL algorithms for practical large buildings.	Existing multi-agent approaches are still not fully scalable for all practical large-building problems, so further algorithmic development is needed.

Table 27: One-line MRQ1 answer for Yu et al. (2021), after recoding.

survey_id	MRQ1 answer
YuEtAl2021	Yu et al. (2021) present a capability-driven algorithm narrative rather than a single dominant winner: value-based DRL is treated as a representative discrete-action baseline, policy-based and actor–critic-style DRL are emphasized for continuous control and stable learning, model-based DRL is strongly highlighted for sample efficiency, and multi-agent DRL is positioned as increasingly important for scalable multi-zone, multi-building, and microgrid-level smart-building energy management.

Table 28: MRQ1 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
WeinbergEtAl2023	3	3	The survey is recommendation-driven rather than dominance-driven. It characterizes value-based methods such as SARSA, Q-learning, and DQN as representative algorithms, but also identifies their limitations for continuous HVAC/BES control. Policy-based and actor–critic methods such as TRPO, PPO, DDPG, and SAC are structurally important because the paper emphasizes sample efficiency, off-policy learning, and continuous-control suitability. Model-based RL is strongly emphasized as a remedy for slow data collection and low sample efficiency in building energy systems. Multi-agent RL is also structurally important because the paper organizes one of its core challenge categories around multi-agent aspects and building clusters. Recommendation receives score 3 because the paper strongly favors sample-efficient directions, especially off-policy actor–critic methods, model-based RL, transfer learning, and multi-agent RL for scalable building-cluster control.

Table 29: MRQ1 method-family characterization scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Weinberg et al. (2023)	2	3	3	3	3

Table 30: MRQ1 method-family recommendation scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Weinberg et al. (2023)	0	2	3	3	3

Table 31: MRQ1 divergence/rationale table for Weinberg et al. (2023), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based algorithms such as DQN, SARSA, and Q-learning are presented as standard representative RL methods and appear in the paper’s taxonomy of popular approaches.	The paper notes that Q-learning, DQN, and SARSA are unsuitable for HVAC control when the action space is large or continuous, so it does not recommend value-based methods as the general solution.
Policy-gradient / policy-based	Policy-based methods such as TRPO, PPO, DDPG, and SAC are characterized as popular RL algorithms and relevant to continuous-control settings.	The strongest recommendation is not for policy-gradient methods as a broad family, but for sample-efficient off-policy and actor–critic-style learning.
Actor–critic	Off-policy actor–critic methods are strongly supported because they can reuse collected building data and improve sample efficiency. SAC and DDPG are highlighted as especially relevant to continuous HVAC/BES control.	The paper still treats these as part of a broader sample-efficiency agenda rather than a universal solution independent of data, dynamics, or deployment context.
Model-based / hybrid	Model-based RL is strongly supported as a remedy for low sample efficiency and slow data collection, and the paper reports examples where model-based methods outperform model-free alternatives in sample efficiency or average reward.	Accurate environment modeling remains difficult, and model-based success depends on useful learned or engineered dynamics models.
Multi-agent / distributed	Multi-agent RL is structurally important because building clusters and multi-agent aspects form a core challenge category. It is treated as a key direction for scalable BES control.	Multi-agent RL is framed mainly as a future-oriented direction for clusters rather than a mature default for all BES settings.
Transfer learning	Transfer learning is highlighted as a prominent strategy for efficient learning in building clusters, either by transferring control policies or plant models between buildings.	Transfer learning is kept as a strategy note rather than a main family column because the main family heatmap focuses on algorithm classes.

Table 32: MRQ1 strategy-level notes for Weinberg et al. (2023), after recoding.

survey_id	note_type	note
WeinbergEtAl2023	Training strategy	Off-policy algorithms are described as attractive for building energy systems because slow sensor sampling makes data expensive, and off-policy methods can reuse all collected data throughout learning.
WeinbergEtAl2023	Algorithm comparison	DDPG is reported as converging much faster than DQN for continuous-action HVAC control.
WeinbergEtAl2023	Algorithm comparison	Off-policy SAC is reported as converging much faster than on-policy counterparts in a simulated data-center setting, with SAC framed as a likely key method for next-generation HVAC control systems.
WeinbergEtAl2023	Future direction	Transfer learning is highlighted as a prominent approach for efficient learning in building clusters, either by transferring control policies or plant models between buildings.

Table 33: One-line MRQ1 answer for Weinberg et al. (2023), after recoding.

survey_id	MRQ1 answer
WeinbergEtAl2023	Weinberg et al. (2023) present a recommendation-driven rather than dominance-driven algorithm narrative: instead of naming one classical RL family as the overall winner, they favor sample-efficient approaches for building energy systems, especially off-policy actor–critic learning, model-based RL, transfer learning, and multi-agent RL for scalable control in building clusters.

7 Latoń et al.

Table 34: MRQ1 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
LatonEtAl2024	3	3	The survey structurally characterizes RL and DRL methods for HEMS across multiple application areas, including demand response, load scheduling, renewable integration, energy storage, electric vehicles, prosumer systems, and distributed home-energy automation. Value-based methods such as Q-learning, DQN, Double DQN, and Dueling DQN are repeatedly represented across HEMS use cases. Actor–critic and modern DRL methods such as PPO, DDPG, A2C, ACKTR, SAC, TD3, and related continuous-control approaches are also widely represented across application groups. Multi-agent and distributed reinforcement learning are structurally important because the review connects them to decentralized control, privacy preservation, scalability, smart-grid interaction, local microgrids, and prosumer-level coordination. Recommendation receives score 3 because the clearest future-oriented direction is toward scalable, privacy-aware, distributed and multi-agent RL/DRL, supported by transfer learning, stronger interoperability, and more real-world deployable HEMS architectures.

Table 35: MRQ1 method-family characterization scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Latoń et al. (2024)	3	0	3	3	0	3

Table 36: MRQ1 method-family recommendation scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Latoń et al. (2024)	2	0	2	2	0	3

Table 37: MRQ1 divergence/rationale table for Latoń et al. (2024), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning and DQN-family methods appear across several HEMS application groups, showing that they remain practically important in demand response, scheduling, renewable-energy integration, storage, and electric-vehicle applications.	The review presents HEMS control as application-dependent and does not elevate value-based methods as the single leading family across all tasks, especially where continuous control, coordination, privacy, and scalability matter.
Actor–critic / modern DRL	PPO, DDPG, ACKTR, A2C, SAC, TD3, and related methods appear across multiple HEMS application areas, especially where richer DRL implementations, scheduling, continuous control, and adaptive optimization are needed.	The paper does not rank these methods as universally best; they are treated as part of a broad, application-dependent algorithm mix.
Deep RL	DRL is central to the survey’s HEMS narrative because it enables more adaptive and complex decision-making than simpler tabular approaches in modern smart-home energy systems.	DRL remains constrained by real-world validation gaps, data requirements, scalability, privacy/security, and interoperability challenges.
Multi-agent / distributed	MARL and distributed control are described as promising for decentralized, privacy-preserving, scalable energy management in smart homes, prosumer systems, local microgrids, and smart-grid-integrated HEMS.	The paper also notes major practical challenges, including limited transferability across heterogeneous buildings, costly retraining, and the need for adaptive learning frameworks.

Table 38: MRQ1 future-direction notes for Latoń et al. (2024), after recoding.

survey_id	note_type	note
LatonEtAl2024	Future direction	The review highlights a strong need for more real-world implementations and case studies, since most existing HEMS RL/DRL studies remain simulation-based.
LatonEtAl2024	Future direction	Transfer learning is presented as a promising way to improve the adaptability of RL and MARL across buildings and systems with limited data.
LatonEtAl2024	Future direction	Security, privacy, scalability, interoperability, and communication standards are emphasized as major challenges that future RL/DRL-based HEMS research must address.

Table 39: One-line MRQ1 answer for Latoń et al. (2024), after recoding.

survey_id	MRQ1 answer
LatonEtAl2024	Latoń et al. (2024) present a mixed, application-dependent DRL narrative for HEMS rather than a single dominant family: both value-based and actor–critic/modern DRL methods are broadly represented across use cases, while the clearest future-oriented recommendation is toward multi-agent and distributed RL, transfer learning, and more scalable, secure, privacy-aware, and real-world deployable energy-management systems.

8 Fu et al.

Table 40: MRQ1 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
FuEtA12022	3	3	The survey provides one of the clearest algorithm-family narratives in the corpus. Value-based algorithms are structurally dominant in the reviewed literature, with Q-learning described as the most classical and most commonly used method, and DQN-like variants extending value-based RL to larger or continuous state spaces with discrete actions. Policy-search methods are explicitly discussed but are less common and less preferred because of policy-evaluation cost. Actor–critic methods are structurally important because they combine value-based and policy-based advantages and are explicitly prioritized for practical problems with large or continuous action spaces. Multi-agent reinforcement learning is also strongly future-oriented because the paper recommends it for complex control scales, multiple control objects, building clusters, and multi-building control. Recommendation receives score 3 because actor–critic and MARL are explicitly recommended for complex, continuous, and multi-object building-control problems, while value-based methods remain conditionally recommended for simpler small/discrete settings.

Table 41: MRQ1 method-family characterization scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Fu et al. (2022)	3	2	3	3	0	3

Table 42: MRQ1 method-family recommendation scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Fu et al. (2022)	2	1	3	3	0	3

Table 43: MRQ1 divergence/rationale table for Fu et al. (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based methods dominate the reviewed literature. Q-learning is simple, easy to deploy, and preferred for small/discrete problems, while DQN-like methods extend the family to large or continuous state spaces with discrete actions.	Value-based methods are weak for continuous action spaces, and discretization can lead to suboptimal control policies compared with policy-based or actor–critic methods.
Policy-search / policy-based	Policy-search methods can handle large or continuous action spaces, and PPO is presented as relatively stable.	The survey says policy-search methods are less used, more costly in policy evaluation, and not the first choice compared with actor–critic methods.
Actor–critic	Actor–critic methods combine the advantages of value-based and policy-based approaches, are more efficient for practical applications, and are explicitly prioritized for complex control problems with large or continuous action spaces. DDPG is highlighted for continuous actions, while A2C/A3C are linked to faster training.	Actor–critic methods are more difficult to implement than simpler value-based methods and are not presented as necessary for small or simple discrete problems.
Deep RL	Deep reinforcement learning is presented as more appropriate than classical RL for complex building-control problems.	DRL still inherits practical limitations such as training time, safety, and implementation difficulty.
Multi-agent / distributed	Multi-agent reinforcement learning is explicitly recommended for building clusters and multiple control objects, and is reported to have better efficiency and performance than single-agent approaches in complex systems.	MARL is framed mainly as a future direction needing more recent research and application before it becomes a mature default.

Table 44: MRQ1 future-direction notes for Fu et al. (2022), after recoding.

survey_id	note_type	note
FuEtA12022	Future direction	Deep reinforcement learning is explicitly presented as more appropriate than existing RL methods for solving more complex building-control problems.
FuEtA12022	Future direction	Multi-agent reinforcement learning is highlighted as having better control efficiency and performance than single-agent RL for complex building-control problems.
FuEtA12022	Future direction	Reinforcement learning combined with occupant behavior is identified as a promising future research direction.

Table 45: One-line MRQ1 answer for Fu et al. (2022), after recoding.

survey_id	MRQ1 answer
FuEtA12022	Fu et al. (2022) present one of the clearest algorithm-narrative shifts in the corpus: value-based RL still dominates the reviewed intelligent-building literature, especially through Q-learning, but actor–critic methods are explicitly prioritized for large or continuous action spaces, deep RL is favored for more complex control problems, and multi-agent RL is recommended for complex multi-object and building-cluster control.

Table 46: MRQ1 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
ZhangEtAl2022	3	3	The survey frames MRQ1 primarily as a control-design and method-selection problem rather than a simple algorithm-family ranking. Classical value-based methods such as Q-learning and SARSA are discussed as part of the RL foundation for BEMS, while actor–critic methods are included as relevant advanced RL approaches. Deep RL is structurally important because it extends RL to complex and high-dimensional BEMS settings. The strongest characterization and recommendation are for model-based and hybrid RL–MPC approaches: the paper argues that MPC should be preferred when transition dynamics are well understood, while RL becomes attractive when environments are difficult to model, and hybrid/model-based RL can combine RL adaptability with MPC-style planning, data efficiency, safety, reliability, and robustness. Recommendation receives score 3 because RL+MPC/model-based RL is the clearest future-oriented direction in the survey.

Table 47: MRQ1 method-family characterization scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Zhang et al. (2022)	2	0	2	3	3	0

Table 48: MRQ1 method-family recommendation scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-search / policy-based	Actor–critic	Deep RL	Model-based / hybrid	Multi-agent / distributed
Zhang et al. (2022)	0	0	1	2	3	0

Table 49: MRQ1 divergence/rationale table for Zhang et al. (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Value-based methods provide the classical RL foundation in BEMS, with Q-learning and SARSA serving as representative baseline methods under unknown dynamics.	Classical value-based methods are limited in high-dimensional and continuous-control settings, so the paper does not present them as the main practical future direction.
Actor–critic	Actor–critic methods are included as a core RL family and are discussed in BEMS applications, indicating relevance for more advanced control settings.	Actor–critic is not singled out as the dominant or most recommended family; support is weaker than for hybrid/model-based approaches.
Deep RL	Deep RL is presented as necessary for extending RL to more complex and high-dimensional BEMS problems and as an important development beyond classical RL.	Deep RL remains constrained by data-efficiency, safety, robustness, and generalization challenges, which the paper treats as major practical barriers.
Model-based / hybrid RL	Model-based and hybrid RL–MPC approaches receive the strongest support because they can combine RL adapt-ability with MPC-style planning and improve data efficiency, safety, reliability, and robustness.	These approaches still depend on useful models, predictive structure, or simulator support, so their applicability is conditional on model availability and problem setup.

Table 50: MRQ1 future-direction notes for Zhang et al. (2022), after recoding.

survey_id	note_type	note
ZhangEtAl2022	Recommendation logic	The paper argues that if transition dynamics are well understood, MPC should be preferred; if the environment is difficult to model, RL becomes attractive.
ZhangEtAl2022	Future direction	The survey highlights hybrid RL + MPC approaches as a promising compromise between reliability and practicality in real-time BEMS.
ZhangEtAl2022	Future direction	Data efficiency, safety, robustness, and generalization are emphasized as central reasons to move toward model-based and hybrid RL formulations.

Table 51: One-line MRQ1 answer for Zhang et al. (2022), after recoding.

survey_id	MRQ1 answer
ZhangEtAl2022	Zhang et al. (2022) frame MRQ1 primarily as a control-design choice rather than a family-dominance question: instead of naming one RL family as state of the art, the survey recommends choosing among classical RL, deep RL, and especially hybrid/model-based RL according to model availability, data efficiency, safety, reliability, and robustness requirements, with the strongest support going to RL+MPC and model-based RL directions.

Table 52: MRQ1 recoded audit table for Al-Ani and Das (2022). Scores use the shared 0–3 evidence-intensity scale.

survey__id	Method-family characterization	Method-family recommendation	Recoded evidence justification
AlAniDas2022	3	3	The survey structurally characterizes deep reinforcement-learning families for HEMS and shows that the field had already diversified beyond classical RL. Deep value-based methods, especially DQN and DDQN, are a major discrete-action strand in the reviewed HEMS literature. Policy-based methods such as PPO and TRPO are present but thinly represented. Actor–critic methods are the strongest family in the survey’s deep-learning landscape, with DDPG, SAC, A2C, A3C, and TD3 listed as representative algorithms and actor–critic described as preferred over deep policy learning. Recommendation receives score 3 because the clearest method-family preference is toward actor–critic approaches, while value-based methods remain conditionally important for discrete-action HEMS settings.

Table 53: MRQ1 method-family characterization scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-based	Actor–critic
Al-Ani and Das (2022)	3	2	3

Table 54: MRQ1 method-family recommendation scores for Al-Ani and Das (2022), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-based	Actor–critic
Al-Ani and Das (2022)	2	0	3

Table 55: MRQ1 divergence/rationale table for Al-Ani and Das (2022), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	DQN is the most-used deep algorithm in the reviewed HEMS literature, and DDQN is also popular, making deep value-based methods the standard discrete-action deep-RL option in this survey.	The paper does not elevate value-based methods as the overall preferred deep-learning family; its stronger preference shifts toward actor–critic methods.
Policy-based	PPO and TRPO appear in the HEMS literature, so the family is not absent and receives method-family characterization.	The paper indicates that only PPO and TRPO have been used so far in HEMS and treats deep policy learning as much less preferred than actor–critic methods.
Actor–critic	Actor–critic is presented as the preferred deep-learning approach relative to deep policy learning and as the largest deep-learning family in the survey, with broad coverage through DDPG, SAC, A2C, A3C, and TD3.	The paper’s recommendation is comparative rather than universal; it prefers actor–critic over deep policy learning, but does not claim that it solves all HEMS control problems without qualification.

Table 56: MRQ1 future-direction notes for Al-Ani and Das (2022), after recoding.

survey__id	note__type	note
AlAniDas2022	Family-use pattern	The paper shows that HEMS deep RL had already diversified beyond classical RL, but the strongest concentration was in DQN/DDQN and actor–critic methods rather than in pure deep policy learning.
AlAniDas2022	Recommendation logic	The paper’s clearest recommendation is comparative: actor–critic is preferred over deep policy learning in the reviewed HEMS literature.
AlAniDas2022	Scope note	The review is HEMS-specific, so its MRQ1 contribution is strongest for home energy management rather than the entire building-control literature.

Table 57: One-line MRQ1 answer for Al-Ani and Das (2022), after recoding.

survey__id	MRQ1 answer
AlAniDas2022	Al-Ani and Das (2022) show that HEMS-focused deep RL had already moved beyond simple value-based methods into a mixed deep-RL landscape, but with a clear hierarchy: DQN/DDQN remain the main deep value-based strand, pure deep policy learning is still very limited, and actor–critic emerges as the preferred deep-learning family in practice.

11 Mason and Grijalva

Table 58: MRQ1 recoded audit table for Mason and Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
MasonGrijalva2019	3	3	The survey captures an early-stage reinforcement-learning narrative in building energy management. Value-based methods, especially Q-learning and its variants, are structurally dominant in the reviewed HVAC and home-energy-management applications. Actor–critic methods are present through several applications, including HVAC, water heaters, and HEMS, but they are not treated as the dominant or preferred family. Multi-agent and distributed reinforcement learning are structurally important because the paper discusses them in smart-home, community, and grid-related settings and explicitly recommends scaling multi-agent RL to larger communities of homes. Recommendation receives score 3 because the clearest future-oriented family-level recommendation is toward larger-scale multi-agent RL, while value-based methods remain conditionally supported as the historical practical workhorse.

Table 59: MRQ1 method-family characterization scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor–critic	Multi-agent / distributed
Mason and Grijalva (2019)	3	2	3

Table 60: MRQ1 method-family recommendation scores for Mason and Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor–critic	Multi-agent / distributed
Mason and Grijalva (2019)	2	0	3

Table 61: MRQ1 divergence/rationale table for Mason and Grijalva (2019), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning is repeatedly described as the most common method in HVAC and home energy management, and many reviewed studies report practical savings with Q-learning and its variants.	The future-directions section shifts attention away from purely classical RL toward deeper function approximation and more complex methods for larger state–action spaces.
Actor–critic	Actor–critic appears in several successful applications, including HVAC, water heaters, and HEMS, and is part of the broader reviewed algorithm mix.	The paper does not elevate actor–critic as the dominant or preferred family across the survey.
Multi-agent / distributed	Multi-agent RL is described as commonly used for smart-home/community and grid-integration problems, and the future-directions section explicitly recommends scaling multi-agent RL to larger communities of homes.	It is mainly associated with multi-home and distributed-control settings, not presented as the default solution for all building-energy-management tasks.

Table 62: MRQ1 future-direction notes for Mason and Grijalva (2019), after recoding.

survey_id	note_type	note
MasonGrijalva2019	Future direction	The paper identifies a strong move toward deep RL methods for larger state–action spaces and more complex control tasks.
MasonGrijalva2019	Future direction	The paper explicitly proposes future work on multi-task RL, transfer learning, multi-objective RL, and meta-RL for building energy management.
MasonGrijalva2019	Future direction	The paper recommends extending multi-agent RL to larger communities of homes and studying scalability and cooperation in decentralized settings.

Table 63: One-line MRQ1 answer for Mason and Grijalva (2019), after recoding.

survey_id	MRQ1 answer
MasonGrijalva2019	Mason and Grijalva (2019) capture an early-stage reinforcement-learning narrative in building energy management: classical value-based RL, especially Q-learning, still dominates practical applications, while future work is expected to move toward deeper function-approximation methods and larger-scale multi-agent control.

12 Huynh et al.

Table 64: MRQ1 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey__id	Method-family characterization	Method-family recommendation	Recoded evidence justification
HuynhEtAl2025	3	3	This is a specialized MORL-focused survey, so MRQ1 should be coded around multi-objective reinforcement-learning design rather than only standard single-objective RL families. Traditional RL is discussed as an important baseline, but is treated as insufficient for smart-building settings with conflicting objectives and changing priorities. MORL is the structuring method family of the survey because the paper centers vector-valued rewards, MOMDPs, Pareto reasoning, scalarization, hybrid MORL, and adaptive control under multiple objectives. Single-policy strategies are structurally discussed through weighted sums, scalarization, constraint-based methods, and SER-style approaches, while multi-policy and adaptive strategies are structurally important because the paper emphasizes Pareto-front methods, preference-independent methods, adaptive policies, occupant feedback, and flexibility under changing priorities. Recommendation receives score 3 because the paper clearly recommends MORL over traditional RL for conflicting smart-building objectives and gives the strongest future-oriented support to adaptive and multi-policy approaches.

Table 65: MRQ1 method-family characterization scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Traditional RL	MORL	Single-policy	Multi-policy / adaptive
Huynh et al. (2025)	2	3	3	3

Table 66: MRQ1 method-family recommendation scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Traditional RL	MORL	Single-policy	Multi-policy / adaptive
Huynh et al. (2025)	0	3	2	3

Table 67: MRQ1 divergence/rationale table for Huynh et al. (2025), after recoding.

Category	Why the survey supports it	Why the survey is cautious / limited
Traditional RL	Traditional RL provides the baseline RL framing and can be extended through reward shaping or scalarization.	The paper argues that single-objective RL is insufficient for smart buildings with multiple conflicting goals, occupant preferences, and changing priorities.
MORL	MORL is better suited to balancing energy, cost, comfort, indoor air quality, and other competing objectives. It improves adaptability and supports more effective decision-making in complex smart-building environments.	MORL is more complex and computationally demanding, which may hinder real-time implementation in large-scale buildings.
Single-policy	Single-policy methods are useful when preferences are known and fixed, and they provide simpler implementations through scalarization, weighted sums, or constraints.	They are less flexible because they produce a single optimized outcome and cannot represent a broader set of trade-offs under changing preferences.
Multi-policy / adaptive	Multi-policy and adaptive approaches are better suited to dynamic environments and changing priorities. They support Pareto-front reasoning, preference-independent decision-making, occupant feedback, and more flexible control.	They are harder to compute, more complex to manage, and depend on robust preference handling and deployment-ready control logic.

Table 68: MRQ1 objective-frequency notes for Huynh et al. (2025).

Objective	Frequency
Energy consumption	33
Energy cost	33
Thermal comfort	33
User comfort	18
Indoor air quality	6
Peak shaving	5
Indoor temperature	5

Table 69: One-line MRQ1 answer for Huynh et al. (2025), after recoding.

survey__id	MRQ1 answer
HuynhEtAl2025	Huynh et al. (2025) show a late-stage shift in RL survey discourse: rather than asking only which standard single-objective RL family is best, the survey argues that smart-building control increasingly requires MORL frameworks that can balance conflicting objectives through adaptive and often multi-policy strategies, with growing emphasis on occupant feedback, flexibility, and real-world deployability.

13 Kannari et al.

Table 70: MRQ1 inclusion-status table for Kannari et al. (2025).

survey_id	Include in MRQ1 family heatmap?	Use in MRQ1 qualitative discussion?	Reason
KannariEtAl2025	No	Yes	The paper focuses on real-building implementation challenges, digital twins, safety, scalability, transparency, privacy, governance, and ethics rather than comparing or recommending RL method families such as value-based, policy-based, actor–critic, model-based, offline, or multi-agent RL.

Table 71: MRQ1 qualitative contribution table for Kannari et al. (2025).

survey_id	year	What it contributes to MRQ1	What it does not contribute
KannariEtAl2025	2025	Shows that later review discourse shifts away from algorithm-family ranking toward deployment-oriented evaluation criteria, especially safety, scalability, digital-twin support, data sharing, transparency, privacy, governance, and RL, model-based RL, multi-agent RL, or other standard RL families.	Does not provide a family-level ranking or recommendation among value-based, policy-based, actor–critic, offline

Table 72: MRQ1 rationale-note table for Kannari et al. (2025).

survey_id	note_type	note
KannariEtAl2025	Rationale shift	The paper shows that recommendation logic has shifted toward safety, robustness, and long-term reliability of adaptive solutions in real buildings.
KannariEtAl2025	Rationale shift	The paper emphasizes scalability and wider practical applicability across building contexts as major criteria for judging RL-based control.
KannariEtAl2025	Rationale shift	The paper treats digital twins as important support for RL development, pretraining, deployment, and operation in real buildings.
KannariEtAl2025	Rationale shift	The paper highlights data sharing, transparency, privacy, ethics, governance, accountability, and human oversight as increasingly important criteria in real-building AI/RL deployment.

Table 73: One-line MRQ1 note for Kannari et al. (2025).

survey_id	MRQ1 note
KannariEtAl2025	Kannari et al. (2025) is not a core source for MRQ1 method-family coding, but it is useful for showing that later review discourse evaluates RL control increasingly through deployment-oriented rationales such as safety, scalability, digital-twin support, data sharing, transparency, privacy, governance, and ethics rather than through algorithm-family ranking alone.

14 Chatterjee and Khovalyg

Table 74: MRQ1 recoded audit table for Chatterjee and Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
ChatterjeeKhovalyg2023	3	3	The survey structurally characterizes RL algorithms for HVAC and dynamic indoor-environment control. Value-based methods, especially Q-learning and DQN-style extensions, are presented as the historical workhorse of RL-based HVAC control, with Q-learning described as the most explored approach in the reviewed literature. Actor–critic and advanced DRL methods are also structurally important because the paper links DDPG, A3C, TD3, SAC, PPO, TRPO, and related methods to continuous-control capability, high-dimensional control, sample efficiency, stability, and dynamic indoor-environment applications. Recommendation receives score 3 because the paper’s strongest method-family recommendation shifts toward continuous-control actor–critic DRL, especially DDPG and A3C, with TD3 and SAC discussed as stronger alternatives when stability, sample efficiency, or policy expressiveness are critical.

Table 75: MRQ1 method-family characterization scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor–critic / advanced DRL
Chatterjee and Khovalyg (2023)	3	3

Table 76: MRQ1 method-family recommendation scores for Chatterjee and Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Value-based	Actor–critic / advanced DRL
Chatterjee and Khovalyg (2023)	2	3

Table 77: MRQ1 divergence/rationale table for Chatterjee and Khovalyg (2023), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning is the most explored algorithm in the reviewed HVAC literature and remains the main historical workhorse. DQN-style extensions also appear repeatedly across the surveyed studies.	Q-learning is practical mainly for simpler or minimal environments, and DQN-style methods still face action-space and discretization limits that motivate a move toward continuous-control actor-critic methods.
Actor-critic / advanced DRL	DDPG is emphasized as a strong choice for dynamic indoor-environment control because it supports continuous action spaces, stabilizes learning through the actor-critic structure, uses off-policy learning, and can operate in high-dimensional settings. A3C, TD3, and SAC are also highlighted as strong alternatives when sample efficiency, stability, or multi-modal policy representation are especially important.	These methods can be more complex, more data-intensive, slower to train, or more computationally expensive than simpler value-based approaches, so the paper presents the choice as application-dependent rather than universally settled.

Table 78: MRQ1 future-direction notes for Chatterjee and Khovalyg (2023), after recoding.

survey_id	note_type	note
ChatterjeeKhovalyg2023	Specialization note	The paper is specialized to dynamic indoor-environment control with triple objectives: energy, comfort, and health.
ChatterjeeKhovalyg2023	Future direction	The paper argues that more studies should bring dynamic indoor-environment design and RL controls under the same framework.
ChatterjeeKhovalyg2023	Deployment gap	The paper notes that RL controllers are still largely in the research-and-development stage, with only a minority of reviewed controllers implemented and tested in real buildings.
ChatterjeeKhovalyg2023	Barrier note	The major obstacles highlighted are time-consuming and data-intensive training, control security, transfer/generalization, and the need for an open, interoperable virtual testbed.

Table 79: One-line MRQ1 answer for Chatterjee and Khovalyg (2023), after recoding.

survey_id	MRQ1 answer
ChatterjeeKhovalyg2023	Chatterjee and Khovalyg (2023) present a specialized algorithm narrative for dynamic indoor-environment control: Q-learning remains the most explored historical HVAC-control method, but the recommendation logic shifts toward continuous-control actor-critic DRL, with DDPG presented as the primary candidate and A3C, TD3, and SAC highlighted as strong alternatives when stability, sample efficiency, or policy expressiveness are critical.

15 Sierla et al.

Table 80: MRQ1 inclusion-status table for Sierla et al. (2022).

survey_id	Include in MRQ1 family heatmap?	Use in MRQ1 qualitative discussion?	Reason
SierlaEtAl2022	No	Yes	The paper focuses on how reinforcement-learning actions are formulated for HVAC control, including action abstraction, actuator commands, setpoints, single-zone and multi-zone applicability, and indoor-environment targets such as temperature, humidity, and air quality. It does not provide a family-level ranking or recommendation among value-based, policy-based, actor–critic, model-based, offline, or multi-agent reinforcement-learning families.

Table 81: MRQ1 qualitative contribution table for Sierla et al. (2022).

survey_id	year	What it contributes to MRQ1	What it does not contribute
SierlaEtAl2022	2022	Shows that method recommendation cannot be separated from control formulation. The paper shifts attention from asking which RL family is best toward asking whether the RL action representation can be mapped meaningfully to real HVAC actions, actuator commands, setpoint changes, supervisory control layers, and indoor-environment objectives.	Does not provide a family-level ranking or recommendation among standard RL algorithm families such as value-based, policy-gradient, actor–critic, model-based, offline, or multi-agent RL.

Table 82: MRQ1 rationale-note table for Sierla et al. (2022).

survey_id	note_type	note
SierlaEtAl2022	Rationale shift	The paper shows that later RL-HVAC reviews increasingly evaluate methods through deployable action formulation rather than algorithm family alone.
SierlaEtAl2022	Rationale shift	The paper emphasizes whether actions correspond to actuator commands, setpoint adjustments, supervisory decisions, or planning-level control.
SierlaEtAl2022	Rationale shift	The paper highlights that action abstraction affects whether RL control is physically meaningful, scalable to multi-zone buildings, and compatible with real building automation systems.
SierlaEtAl2022	Rationale shift	The paper is therefore useful for qualitative MRQ1 synthesis, even though it should not be included in the main method-family heatmap.

Table 83: One-line MRQ1 note for Sierla et al. (2022).

survey_id	MRQ1 note
SierlaEtAl2022	Sierla et al. (2022) is not a core source for MRQ1 method-family coding, but it is useful for showing that later review discourse evaluates RL methods through formulation-level criteria, especially whether actions can be mapped to realistic HVAC setpoints, actuators, control layers, and indoor-environment targets.

16 Pinthurat et al.

Table 84: MRQ1 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
PinthuratEtAl2024	3	3	The survey structurally characterizes reinforcement-learning methods for smart-home energy management systems with energy storage. Value-based methods such as Q-learning, DQN, and DDQN are central to the reviewed single-agent RL literature. Policy-gradient or policy-based methods such as PPO are discussed, but are less central than value-based and actor–critic methods. Actor–critic and modern DRL methods such as DDPG, TD3, and SAC are structurally visible because they support more complex and continuous control settings. Model-based and hybrid RL are discussed through the model-free/model-based distinction and through the paper’s future emphasis on hybrid RL algorithms. Multi-agent and distributed RL are structurally important because the paper explicitly separates single-agent and multi-agent RL and treats multi-agent methods as promising for coordinated smart homes, storage systems, and multiple controllable devices. Recommendation receives score 3 because the clearest future-oriented recommendations are toward hybrid RL and multi-agent RL for scalable, coordinated, and practically deployable SHEMS with storage.

Table 85: MRQ1 method-family characterization scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic / modern DRL	Model-based / hybrid	Multi-agent / distributed
Pinthurat et al. (2024)	3	2	3	2	3

Table 86: MRQ1 method-family recommendation scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor-critic / modern DRL	Model-based / hybrid	Multi-agent / distributed
Pinthurat et al. (2024)	2	1	2	3	3

Table 87: MRQ1 divergence/rationale table for Pinthurat et al. (2024), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning, DQN, and DDQN are structurally represented in the single-agent RL literature for SHEMS with storage and remain important practical methods for discrete decision settings.	They are less suitable as the main future direction when control requires continuous decisions, coordination across devices, storage management, and adaptation to changing user, weather, and grid conditions.
Policy-gradient / policy-based	PPO and related policy-based methods are included in the algorithm review and are relevant to more flexible policy learning.	They are not presented as the dominant or clearest future family in the paper; support remains weaker than for actor-critic, hybrid, and multi-agent methods.
Actor-critic / modern DRL	DDPG, TD3, SAC, and related modern DRL methods are relevant for more complex and continuous-control SHEMS settings, especially when storage and flexible devices are included.	The paper does not make actor-critic alone the strongest future recommendation; it places stronger future emphasis on hybrid and multi-agent RL.
Model-based / hybrid	Hybrid RL is explicitly highlighted as a promising future direction because combining complementary learning or control mechanisms can improve robustness, scalability, and practical applicability.	The paper does not treat model-based/hybrid methods as the dominant current family in the reviewed literature.
Multi-agent / distributed	Multi-agent RL is strongly supported for coordinated smart-home communities, multiple energy resources, storage systems, and distributed controllable devices. It is one of the clearest future-oriented directions.	Multi-agent methods add coordination, communication, scalability, and privacy/security challenges, so the recommendation is future-oriented rather than a settled universal default.

Table 88: MRQ1 future-direction notes for Pinthurat et al. (2024), after recoding.

survey_id	note_type	note
PinthuratEtAl2024	Future direction	Hybrid RL algorithms are identified as a promising future direction for RL-based smart-home energy management with storage.
PinthuratEtAl2024	Future direction	Multi-agent RL is identified as a major future direction for coordinating multiple devices, storage systems, and smart-home/community-level energy management.
PinthuratEtAl2024	Future direction	Future RL methods should better support demand-side management, demand response, user preferences, privacy/security, and real-world testing.

Table 89: One-line MRQ1 answer for Pinthurat et al. (2024), after recoding.

survey_id	MRQ1 answer
PinthuratEtAl2024	Pinthurat et al. (2024) characterize RL for smart-home energy management with storage through both single-agent and multi-agent algorithm families: value-based and actor-critic methods remain structurally important, while the strongest future-oriented recommendations point toward hybrid RL and multi-agent RL for scalable, coordinated, storage-aware, and real-world deployable SHEMS.

Table 90: MRQ1 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.			
survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
MichailidisEtAl2025	3	3	The survey structurally classifies reinforcement-learning studies for renewable-energy-integrated BEMS by algorithmic approach, agent framework, value-action network, reward design, baseline controls, RES type, BEMS type, and building typology. Value-based methods such as Q-learning, DQN, DDQN, and D3QN are repeatedly visible in the reviewed literature. Policy-gradient or policy-based methods are discussed but are less prevalent. Actor–critic methods such as TD3, DDPG, and SAC are strongly characterized because they are suitable for continuous, high-dimensional, and adaptive RES-integrated building control. Model-based and hybrid RL are structurally important because hybrid RL is described as an emerging and rapidly growing niche that combines RL with methods such as fuzzy logic, ANN, MPC, and RBC. Multi-agent and distributed RL are also structurally important because the paper evaluates the role of multi-agent and hybrid methods in improving adaptability, coordination, and occupant comfort across RES-integrated BEMS. Recommendation receives score 3 because the strongest future-oriented directions are actor–critic, hybrid/model-based, and multi-agent approaches for scalable, safe, transferable, and coordinated RES-integrated building control.

Table 91: MRQ1 method-family characterization scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Michailidis et al. (2025)	3	2	3	3	3

Table 92: MRQ1 method-family recommendation scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Value-based	Policy-gradient / policy-based	Actor–critic	Model-based / hybrid	Multi-agent / distributed
Michailidis et al. (2025)	2	2	3	3	3

Table 93: MRQ1 divergence/rationale table for Michailidis et al. (2025), after recoding.

Family	Why the survey supports it	Why the survey is cautious / limited
Value-based	Q-learning, DQN, DDQN, D3QN, and other value-action-network methods are repeatedly tabulated across RES-integrated BEMS studies, showing that value-based RL remains important in the historical literature.	Value-based methods are not the strongest future direction for high-dimensional, continuous, multi-objective, and RES-integrated building-control problems.
Policy-gradient / policy-based	Policy-based methods are discussed as useful for continuous and stochastic control settings and as part of the broader RL algorithm landscape.	The paper treats this family as lower-prevalence and less central than actor–critic, hybrid, and multi-agent approaches.
Actor–critic	Actor–critic methods such as TD3, DDPG, and SAC are strongly supported for continuous, high-dimensional, adaptive, and RES-integrated BEMS control.	Their effectiveness depends on problem formulation, training stability, and deployment conditions, so they are not presented as a complete solution by themselves.
Model-based / hybrid	Hybrid RL is presented as an emerging and rapidly growing direction that can combine RL with FLC, ANN, MPC, RBC, or other control methods to improve performance and deployment relevance.	Hybrid methods increase system-design complexity and depend on suitable integration between learning and model/control components.
Multi-agent / distributed	Multi-agent RL is strongly supported for RES-integrated BEMS, districts, microgrids, and coordinated control of interacting energy assets.	MARL introduces communication, coordination, scalability, and stability challenges that still require further development.

Table 94: MRQ1 future-direction notes for Michailidis et al. (2025), after recoding.

survey_id	note_type	note
MichailidisEtAl2025	Future direction	Actor–critic methods such as TD3, DDPG, and SAC are future-relevant for continuous and high-dimensional RES-integrated building-control problems.
MichailidisEtAl2025	Future direction	Hybrid RL is identified as an emerging and rapidly growing direction for combining RL with established control and prediction methods.
MichailidisEtAl2025	Future direction	Multi-agent and hierarchical approaches are important future directions for coordinated RES-integrated BEMS, districts, and microgrid-level control.
MichailidisEtAl2025	Future direction	Future work should improve transferability, robustness, safety/compliance, privacy-preserving distributed learning, and real-world deployment.

Table 95: One-line MRQ1 answer for Michailidis et al. (2025), after recoding.

survey_id	MRQ1 answer
MichailidisEtAl2025	Michailidis et al. (2025) characterize RL for renewable-energy-integrated BEMS through a broad algorithmic taxonomy in which value-based methods remain historically important, but the strongest future-oriented recommendations shift toward actor–critic, hybrid/model-based, and multi-agent approaches for continuous, high-dimensional, coordinated, and RES-integrated building-control problems.

Table 96: MRQ1 recoded audit table for Li et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey__id	Method-family characterization	Method-family recommendation	Recoded evidence justification
Li2025	2	2	Characterization receives score 2 because Section 2.3 and Figure 3 form a dedicated classification subsection that sorts reinforcement learning along eight dimensions, including environmental model utilization and strategy optimization. The classification is not applied afterwards: Section 3 is organized by application and system life cycle, and Tables 1 and 2 record per-study algorithm names without grouping them into families. Value-based, policy-based and actor–critic each appear once in the whole survey, inside the same sentence of the Section 2.3 enumeration. Recommendation receives score 2 because the survey develops advocacy for one family only, multi-agent reinforcement learning, through the dedicated future-direction subsection in Section 5.2, and does not compare or rank the classical families against one another.

Table 97: MRQ1 method-family characterization scores for Li et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Li et al. (2025)	1	1	1	1	3

Table 98: MRQ1 method-family recommendation scores for Li et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Li et al. (2025)	1	0	0	0	3

Table 99: MRQ1 divergence/rationale table for Li et al. (2025), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Multi-agent / distributed	Multi-agent reinforcement learning is the one family the survey develops in its own right. It appears as two of the eight classification dimensions in Figure 3, carries a dedicated column in the chronological study table, and receives a future-direction subsection covering deep multi-agent variants, centralized training with decentralized execution, attention-based communication protocols, and coordination of distributed energy assets.	Concurrent learning across agents is presented as a source of training instability, and scalability is named as a limitation. The survey treats these as open problems to be solved rather than as reasons to avoid the family.
Model-based / hybrid	Model-based and hybrid methods appear as the first classification dimension in Figure 3, with Dyna-Q and model predictive control given as model-based examples and model-based value expansion as a hybrid example. Hybrid approaches combining hierarchical reinforcement learning with model predictive control are named once as a future direction.	The survey’s substantive references to model-based approaches are comparisons against conventional model-based control as a baseline, where reinforcement learning is credited with avoiding the need for precise system models. The family itself is not developed as a reinforcement learning design choice.
Value-based	Value-based methods are named in the Section 2.3 classification with Q-learning given as the example, and learning and deep Q-network variants appear as per-study algorithm labels in both study tables.	The family label is never used again after Section 2.3, and the per-study algorithm entries are not aggregated or analyzed at family level.
Policy-gradient / policy-based	Policy-based methods are named in the Section 2.3 classification with REINFORCE given as the example.	No further treatment appears anywhere in the survey, and no study is characterized as belonging to this family.
Actor–critic	Actor–critic methods are named in the Section 2.3 classification with A2C and TRPO given as examples.	Proximal policy optimization, deep deterministic policy gradient, soft actor–critic and twin delayed deep deterministic policy gradient appear as per-study algorithm labels, but the survey never identifies them as actor–critic methods or discusses the family.

Table 100: One-line MRQ1 answer for Li et al. (2025), after recoding.

survey__id	MRQ1 answer
Li2025	Li et al. (2025) present a method-family taxonomy in a dedicated classification subsection but then organize the review by application and system life cycle rather than by algorithm family, so value-based, policy-based, actor–critic and model-based methods are named once and not developed; multi-agent reinforcement learning is the single family that reaches a structuring role, through a dedicated study-table column and a future-direction subsection.

Table 101: MRQ1 recoded audit table for Lei et al. (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
LeiEtAl2026	3	3	Characterization receives score 3 because Section 3 states that it categorizes existing deep reinforcement learning approaches into six distinct classes based on algorithmic structure and contribution to resilience, and then gives each class its own subsection and a place in Figure 4. Value-based methods, actor–critic frameworks and multi-agent reinforcement learning are three of the six; the remaining three are safe, offline and hierarchical reinforcement learning, which fall outside the MRQ1 canonical dimension set. Recommendation receives score 3 because each subsection closes with an explicit suitability statement, and Section 3.8 consolidates these into a paradigm-to-task mapping organized along decision urgency and constraint strictness, supported by Figure 5 and by the comparison of characteristics, strengths, limitations and suitability in Table 6.

Table 102: MRQ1 method-family characterization scores for Lei et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Lei et al. (2026)	1	3	2	3	3

Table 103: MRQ1 method-family recommendation scores for Lei et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Lei et al. (2026)	0	3	0	3	3

Table 104: MRQ1 divergence/rationale table for Lei et al. (2026), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Actor–critic	Actor–critic frameworks are named the most natural choice for continuous building multi-energy system control, particularly where fine-grained regulation of energy flows, thermal dynamics and storage operation is required. The survey traces the family from the policy-gradient foundation through A3C, DPG and DDPG to PPO and SAC, and the paradigm-to-task mapping assigns it to real-time continuous control with high-dimensional action spaces.	Actor–critic methods can be unstable in sparse-reward resilience settings and do not guarantee constraint satisfaction unless additional safety mechanisms are introduced. Continuous policy updates may extrapolate poorly to unseen extreme-event states, which limits reliability under high-impact low-probability conditions.
Value-based	Value-based methods are presented as well suited to discrete decisions such as load-shedding priority ordering and islanding mode selection. Distributional variants including C51, QR-DQN, IQN and Rainbow are credited with supporting risk-sensitive decision-making in resilience-oriented control.	Applicability is limited by the discrete action assumption, because many practical control variables in these systems, including heat pump setpoints, electrolyzer power levels and storage rates, are continuous.
Multi-agent / distributed	Multi-agent reinforcement learning is presented as well suited to distributed control because it improves fault tolerance and supports coordinated reconfiguration during disruptions, and the paradigm-to-task mapping assigns it to emergency response scenarios requiring coordination among multiple subsystems.	Training is often unstable under high-impact low-probability conditions because the environment becomes non-stationary as multiple agents update simultaneously. Strong physical coupling among energy carriers makes credit assignment difficult and joint policy convergence less reliable, and centralized training with decentralized execution and value decomposition mitigate but do not remove these problems.
Policy-gradient / policy-based	Policy-gradient methods are credited with providing the theoretical foundation of actor–critic algorithms, through the result that policy updates can be guided by value estimates without explicitly computing full returns.	The family is positioned only as a predecessor. Actor–critic frameworks are described as generally more stable than pure policy-gradient methods, and early actor–critic work is credited with improving on high-variance Monte Carlo policy gradients. Policy-gradient methods are not one of the six classes and receive no task assignment in the mapping.
Model-based / hybrid	No supporting treatment appears. The family is not one of the six classes and receives no task assignment.	Model-based approaches appear only as a contrast baseline: deep reinforcement learning is introduced as unlike traditional model-based optimization methods that rely on accurate physical modeling and linearization, and the field is described as transitioning from model-based steady-state optimization toward data-driven dynamic resilience control.

Table 105: One-line MRQ1 answer for Lei et al. (2026), after recoding.

survey_id	MRQ1 answer
LeiEtAl2026	Lei et al. (2026) build the survey on an explicit six-class taxonomy of deep reinforcement learning, give each class its own subsection, and close with a paradigm-to-task mapping that assigns value-based methods to discrete switching and load shedding, actor–critic methods to real-time continuous control, and multi-agent reinforcement learning to emergency coordination; three of the six classes (safe, offline and hierarchical reinforcement learning) lie outside the canonical family axis, and policy-gradient methods appear only as the theoretical predecessor of actor–critic.

Table 106: MRQ1 recoded audit table for Mazhar et al. (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family characterization	Method-family recommendation	Recoded evidence justification
MazharEtAl2026	3	3	Characterization receives score 3 because Section 4.1 gives deep Q-networks, proximal policy optimization, trust region policy optimization, and advantage actor–critic methods their own subsections, each closing with an explicit strengths and limitations list, and because the families are carried through two comparative tables and a process figure. Family labels are used in the survey’s own voice: deep Q-networks are named a value-based algorithm and proximal policy optimization a policy-gradient algorithm. Recommendation receives score 3 because Section 4.12 is a dedicated justification subsection that selects deep Q-networks for discrete control in resource-constrained settings and proximal policy optimization for continuous control, and presents the two as complementary.

Table 107: MRQ1 method-family characterization scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Mazhar et al. (2026)	0	3	3	3	1

Table 108: MRQ1 method-family recommendation scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Mazhar et al. (2026)	0	3	3	2	1

Table 109: MRQ1 divergence/rationale table for Mazhar et al. (2026), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Value-based	Deep Q-networks are named a value-based algorithm suited to discrete action spaces, credited with effectiveness in load scheduling and on/off control of heating, ventilation, air conditioning and lighting, and with computational efficiency that suits buildings with limited processing capability. The dedicated justification subsection selects them for discrete control in resource-constrained environments.	Deep Q-networks are said to struggle in complex multi-zone buildings requiring fine-grained control, and performance is said to decline in the continuous state and action spaces common in real-time energy optimization.
Policy-gradient / policy-based	Proximal policy optimization is named a policy-gradient algorithm and is credited with efficiency and stability in continuous control, adjusting actions in small incremental steps in a way suited to nuanced heating, ventilation and air conditioning adjustment and to renewable allocation under fluctuating demand. The justification subsection selects it for continuous control tasks.	Proximal policy optimization is said to require more training time and processing power than deep Q-networks, which makes it less feasible for small-scale buildings, and its performance is said to depend heavily on hyperparameter tuning. Trust region policy optimization is credited with stable updates but slower training, making it less suitable where frequent retraining is needed.
Actor–critic	Advantage actor–critic and its asynchronous variant are credited with combining policy-based and value-based benefits, handling large multi-zone environments under real-time control requirements, and converging faster through asynchronous updates. Two deployments are cited: a national laboratory testbed reporting up to thirty percent reduction in heating, ventilation and air conditioning energy, and a data-centre cooling system reporting forty percent reduction in cooling energy cost.	Both variants are said to require considerable computational resources at scale, to be complex to implement and maintain, potentially requiring specialized hardware, and to need extensive hyperparameter tuning. The dedicated justification subsection selects only deep Q-networks and proximal policy optimization, so actor–critic methods are characterized in full but not carried into the survey’s selection rationale.
Multi-agent / distributed	Multi-agent reinforcement learning appears through a named benchmark environment described as suited to evaluating decentralized models for grid-interactive smart buildings, and adoption of multi-agent approaches is listed once among future research directions.	The family receives no subsection, no strengths and limitations treatment, and no place in the comparative tables or the selection rationale.
Model-based / hybrid	No supporting treatment appears. The term model-based does not occur in the survey.	The hybrid models discussed in Section 4.9.3 are ensemble and sequential predictive forecasting models rather than hybrid reinforcement learning, so they do not bear on this family.

Table 110: One-line MRQ1 answer for Mazhar et al. (2026), after recoding.

survey_id	MRQ1 answer
MazharEtAl2026	Mazhar et al. (2026) organize their treatment of methods around four named algorithms, giving deep Q-networks, proximal policy optimization, trust region policy optimization and advantage actor–critic methods their own subsections with paired strengths and limitations, and then narrow to an explicit selection rationale that pairs value-based methods with discrete control and policy-gradient methods with continuous control; multi-agent reinforcement learning appears only as a benchmark environment and a future-direction bullet, and model-based reinforcement learning is absent.

Table 111: MRQ1 recoded audit table for Mohsenpour and Xing (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Method-family	characterization	Method-family recommendation	Recoded evidence justification
MohsenpourXing2026	3	3	Characterization receives score 3 because every canonical family has its own subsection: value-based, policy-based, actor–critic, multi-agent, and model-free and model-based reinforcement learning. The families are also carried by a taxonomy figure separating value-based, policy-based and actor–critic models, a dedicated actor–critic framework figure, and a summary figure of advantages and disadvantages of reinforcement learning models. Hybrid reinforcement learning receives a full results subsection with a comparative matrix of hybrid combinations across the reviewed studies. Recommendation receives score 3 because the survey states an explicit family-to-task assignment, applying value-based methods to discrete action spaces and preferring actor–critic methods for continuous control, and because hybrid reinforcement learning is advanced as the response to the tabulated limitations of standalone reinforcement learning.	

Table 112: MRQ1 method-family characterization scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Mohsenpour and Xing (2026)	3	3	3	3	3

Table 113: MRQ1 method-family recommendation scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale.

Survey	Model-based / hybrid	Value-based	Policy-gradient / policy-based	Actor–critic	Multi-agent / distributed
Mohsenpour and Xing (2026)	3	3	2	3	2

Table 114: MRQ1 divergence/rationale table for Mohsenpour and Xing (2026), after recoding.

Family	Why the survey supports it	Why the survey is cautious
Model-based / hybrid	Model-based reinforcement learning receives its own subsection and is credited with mitigating real-world deployment difficulties such as data-collection cost and time constraints, and with faster convergence, supported by co-simulation interfaces linking building simulation tools to computational platforms. Hybrid reinforcement learning then receives a dedicated results subsection in which integration with optimization, transfer learning, large language models, deep learning and physics-informed machine learning is presented as the way to overcome the tabulated limitations of standalone reinforcement learning, with reported gains including a genetic-algorithm and deep Q-network combination improving on the standalone algorithm and a transfer-learning soft actor–critic improving energy efficiency across offices.	The survey frames hybrid approaches as a response to unresolved problems rather than a settled solution, since the limitations that motivate them, including sparse and delayed rewards, high-dimensional occupant-energy interactions, weak generalizability, and high training cost in data-scarce settings, are presented as still open.
Value-based	Value-based methods receive their own subsection developing value and action-value functions and the Bellman optimality relation, appear in the taxonomy figure and in the summary of advantages and disadvantages, and are assigned to discrete action spaces in the survey’s family-to-task statement.	Value-based methods are not presented as adequate for continuous control, which is assigned to actor–critic methods instead.
Actor–critic	Actor–critic methods receive their own subsection and their own framework figure, appear in the taxonomy figure and the advantages and disadvantages summary, and are stated to be preferred for continuous control. Several of the hybrid frameworks reviewed are built on soft actor–critic.	The survey does not present actor–critic methods as universally superior, and their role is defined relative to the discrete and continuous split rather than as a general ranking.
Policy-gradient / policy-based	Policy-based methods receive their own subsection, credited with learning the policy directly, faster convergence in high-dimensional or continuous action spaces, the ability to model a probability distribution over actions for more nuanced control, and effectiveness under stochastic uncertainty. They appear in the taxonomy figure and the advantages and disadvantages summary.	The family-to-task statement names only value-based and actor–critic methods, so policy-based methods are characterized in full but are not carried into the survey’s task assignment.
Multi-agent / distributed	Multi-agent reinforcement learning receives its own subsection, framed through Markov games and organized into a three-way taxonomy of fully centralized, fully decentralized, and centralized training with decentralized execution architectures.	The endorsement is internal to the family rather than across families. Fully centralized training is said to introduce high computational complexity and fully decentralized learning to risk non-convergence, with centralized training and decentralized execution presented as the balanced trade-off between scalability and coordination.

Table 115: One-line MRQ1 answer for Mohsenpour and Xing (2026), after recoding.

survey_id	MRQ1 answer
MohsenpourXing2026	Mohsenpour and Xing (2026) give every canonical method family its own subsection and supporting figure, assign value-based methods to discrete action spaces and actor–critic methods to continuous control, and then treat hybrid reinforcement learning as the survey’s organizing argument, presenting integration with optimization, transfer learning, large language models and physics-informed machine learning as the route past the limitations of standalone reinforcement learning.